Quiz 3 Solutions

1. **(a) Output Shape Calculation**
   - Input: 64 × 64 × 1
   - Conv Layer 1 (same padding)
     - Output shape = ⌊(64 + 2×2 - 5)/1 + 1⌋ = 64 × 64 × 16
   - Max Pooling 1
     - Output shape = ⌊64 / 2⌋ × ⌊64 / 2⌋ × 16 = 32 × 32 × 16
   - Conv Layer 2 (valid padding)
     - Output size = ⌊(32 - 3)/1 + 1⌋ = 30 × 30 × 32
   - Max Pooling 2
     - Output = 15 × 15 × 32

Final output shape: 15 × 15 × 32

(b) Trainable Parameters

- Conv Layer 1:
  - Each filter has 5×5×1 = 25 weights + 1 bias
  - Total = (25 + 1) × 16 = 416 parameters
- Conv Layer 2:
  - Each filter has 3×3×16 = 144 weights + 1 bias
  - Total = (144 + 1) × 32 = 4640 parameters

Total Trainable Parameters = 416 + 4640 = 5056

2. (a) Output Shape Calculations

Input: 128 × 128 × 3

Conv Layer 1

- Filter size = 7×7, stride = 1, padding = 'same'
- Output size (same padding) = 128 × 128 × 32

Max Pooling Layer 1

- Pool size = 2×2, stride = 2
- Output = 64 × 64 × 32

Conv Layer 2

- Kernel size = 5×5, padding = 'valid', stride = 1
- Output spatial size = 60×60×64

Max Pooling Layer 2

- Pool size = 2×2, stride = 2
- Output = 30 × 30 × 64

Final Output Shape: 30 × 30 × 64

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(b) Trainable Parameters

Conv Layer 1

- Each filter: 7×7×3 = 147 weights + 1 bias = 148
- Total filters = 32
- Parameters = 148 × 32 = 4736

Conv Layer 2

- Each filter: 5×5×32 = 800 weights + 1 bias = 801
- Total filters = 64
- Parameters = 801 × 64 = 51264

Total Parameters = 4736 + 51264 = 56000

3.

Step 1: Hidden Layer Pre-activation

$$z^{[1]} = W^{[1]} \cdot \mathbf{x} + b^{[1]} = \begin{bmatrix} 0.5 & -1 & 2 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

First row:

$$0.5(1) + (-1)(2) + 2(-1) = 0.5 - 2 - 2 = -3.5$$

Second row:

$$-2(1) + 1(2) + 0 = -2 + 2 = 0 \Rightarrow z^{[1]} = \begin{bmatrix} -3.5 \\ 0 \end{bmatrix}$$

Step 2: tanh Activation

$$a^{[1]} = \tanh(z^{[1]}) = \begin{bmatrix} \tanh(-3.5) \\ \tanh(0) \end{bmatrix} = \begin{bmatrix} \approx -0.998 \\ 0 \end{bmatrix}$$

Step 3: Output Layer

$$z^{[2]} = W^{[2]} \cdot a^{[1]} + b^{[2]} = (1)(-0.998) + (-1)(0) = -0.998$$

Apply sigmoid:

$$a^{[2]} = \sigma(-0.998) \approx \frac{1}{1 + e^{0.998}} \approx 0.269 \Rightarrow \text{Final Output: } \boxed{0.269}$$

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4.

🔹 **Step 1: Hidden Layer Pre-activation**

$$z^{[1]} = W^{[1]} \cdot \mathbf{x} + b^{[1]}$$

Compute:

$$z^{[1]} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

- First neuron: $1(1) + (-1)(-2) + 0(1) = 1 + 2 = 3$
- Second neuron: $2(1) + 0(-2) + (-1)(1) = 2 + 0 - 1 = 1$
- Third neuron: $0(1) + 1(-2) + 1(1) = -2 + 1 = -1$

Add biases:

$$z^{[1]} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$$

🔹 **Step 2: Apply ReLU Activation**

$$a^{[1]} = \text{ReLU}(z^{[1]}) = \begin{bmatrix} \max(0,3) \\ \max(0,2) \\ \max(0,-2) \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$

🔹 **Step 3: Output Layer Pre-activation**

$$z^{[2]} = W^{[2]} \cdot a^{[1]} + b^{[2]} = \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} + 0.5$$

$$z^{[2]} = (2)(3) + (-1)(2) + (1)(0) + 0.5 = 6 - 2 + 0 + 0.5 = 4.5$$

🔹 **Step 4: Apply Sigmoid Activation**

$$\hat{y} = \sigma(z^{[2]}) = \frac{1}{1 + e^{-4.5}} \approx \frac{1}{1 + 0.0111} \approx 0.989$$

✅ **Final Output:** $\boxed{0.989}$